

International Islamic University Chittagong
Center for General Education
Midterm Exam- Spring 2026
Course Code: GEED- 1201
Course Title: Basic Principles of Islam

Time: 1.30 Hours

Full Marks: 30

Answer the following questions
(All questions are of equal value)

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	Define Islam literally and terminologically. Discuss some important characteristics of Islam elaborately.	10	2	Remember
2	Analyze the different the type of <i>Tawhid</i> in the light of <i>Qur'an</i> and Sunnah along with its impact on Muslims' life. Or Make a comparison among all divine books and scriptures mentioning the authenticity and finality of the holy <i>Qur'an</i> with proper evidences.	10	2	Analyze Evaluate
3	What are the different stages of <i>Akirah</i> ? Explain the impact of belief in <i>Akhirah</i> in the individual and collective life of the Muslims	10	2	Remember& Create



International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Spring 2026
Course Code: CSE-1221 Title: Computer Programming 2

Total Marks: 30 Time: 90 minutes

Answer all 3 from the following Questions. Figures in the right-hand margin indicate full marks

Answers must be done sequentially

- #
- 1 a) Define object-oriented programming? Describe the four major characteristics of OOP with short example. Marks COs DL 3 CO1 C1
- b) Compare Structured Programming and Object-Oriented Programming (OOP) in terms of data security and design approach. Why is the "bottom-up" approach considered more suitable for complex systems? 3 CO1 C3

Or, "A class is often called an abstract data type." Justify this statement. Briefly explain how a class differs from a struct regarding default access members.

- c) Analyze the following code. Identify the errors that will prevent it from compiling and rewrite the corrected version. 2 CO1 C2

```
#include <iostream>
using namespace std;
class Teacher {
    int teacher_id;
public:
    Teacher(int id) { teacher_id = id; }
    void show() {
        cout << "ID: " << teacher_id; }
};
```

```
int main() {
    Teacher t1;
    t1.teacher_id = 5001;
    t1.show();
    return 0;
}
```

- d) Define the purpose of the this pointer. Write a constructor for a class Box(int l, int w) where the parameter names are identical to the class member names. 2 CO1 C3
- 2 a) How do you identify the use of parameterized constructor in your code? Is it possible to use both parameterized and non-parameterized constructor at the same program? How and why? Give a concrete example with explanation. 3 CO2 C1

or) What is an "automatic" inline function? When should we use the inline keyword, and when is it better to avoid it? Provide a short C++ example to support your answer

- b) Write the output of the following C++ program:

```
#include <iostream>
using namespace std;
class Simple {
    int id;
public:
    Simple(int i) { id = i; cout << "C" << id << " "; }
    ~Simple() { cout << "D" << id << " "; }
};
int main() {
    Simple s1(1);
    {
        Simple s2(2);
    }
    cout << "End ";
    return 0; }
```

2 CO1 C2

#

c) Write a class named "Product" with private member variables like: *P_id*, *P_name*, *P_company*, *P_price*. 3 CO2 C3

- Create *n* number of products in the main function.
- Take input for each object using *cin* in a member function of class called *set_product*. Show the user a particular product information by taking *P_id* from the user.

d) In your code in 2(c), there is a private variable called *P_price*. Now rewrite the code with a non-member function called "Update_price(double p)" so that it can access the *P_price* variable without any member function of that class. 2 CO2 C3

3 a) Can you overload a function based only on its return type? (e.g., *int func(int x)* vs *double func(int x)*). Explain why the compiler can or cannot distinguish between them. 3 CO2 C2

b) What is a Copy Constructor? Mention three specific scenarios where a copy constructor is automatically invoked by the compiler. 2 CO2 C3

c) Why we must be careful when using default argument in a code where function overloading is applied? Explain with appropriate code. 2 CO2 C2

d) In the following code, *add* function is called two times in main function. Make the required adjustments to ensure that both calls are successful. 3 CO2 C3

```
#include <iostream>
using namespace std;
int add(int i, int j, int k){
    return i+j+k;
}
int main(){
    int i, j, k;
    cin >> i >> j >> k;
    cout << add(i, j, k) << endl;
    cout << add (i, k) << endl;
}
```

or) In the following code, you can see a non-member function called *pri()*. Complete the function so that it can show the value of *id* without changing anything in the class and main function.

```
#include <iostream>
using namespace std;
class A{
    int id;
public:
    A(int a){ id = a; }
    int get_id(){ return id; }
};
```

```
void pri(){
    // write code here...
}
void main(){
    int id;
    cin >> id;
    A ob(id);
    pri(ob);
}
```

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Mid Term Examination

Program: B. Sc. in CSE
 Course Code: MATH-1207
 Time: 1:30 hours



Semester: Spring 2026
 Course Title: Mathematics-II
 Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Please answer the several parts of a question sequentially.
 (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Demonstrate knowledge of geometry and its applications in the real life contexts as well as into complex engineering problems.
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Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- | | | Marks | CLO | DL |
|------------------|---|-------|------|----|
| 1. a) | Transform the equation, $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ inclined at an angle $\tan^{-1}\left(\frac{-4}{3}\right)$ to the original axes. | 5 | CLO1 | U |
| | Or) | | | |
| b) | If any one of the lines $ax^2 + 2hxy + by^2 = 0$ be perpendicular to any one of the lines $px^2 + 2h_1xy + qy^2 = 0$ then prove that, $(ap - bq)^2 + 4(ph + bh_1)(ah_1 + qh) = 0$ | | | |
| a) | Find the value of k so that the following equation may represent pairs of straight lines, $12x^2 - 10xy + 2y^2 + 11x - 5y + k = 0$ | 5 | CLO1 | U |
| | Or) | | | |
| d) | Test the nature of the conic given by the equation,
$10x^2 - 124xy - 10y^2 + 24x + 98y - 79 = 0$ | | | |
| 2. a) | Prove that the straight lines whose direction cosine are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$ are perpendicular if, $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ and parallel if, $\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$ | 5 | CLO1 | U |
| a) | Determine the constant k so that the planes $x - 2y + kz = 0$ and $2x + 5y - z = 0$ are at right angles. Find in that case the plane through the point $(1, -1, -1)$ and perpendicular to both the given planes. | 5 | CLO1 | U |
| 3. a) | Find the shortest distance between the lines $\frac{x-2}{2} = \frac{y-3}{-3} = \frac{z-4}{2}$ and $\frac{x-3}{4} = \frac{y-5}{1} = \frac{z-6}{-3}$ | 5 | CLO1 | U |
| b) | A plane passes a fixed point (a, b, c) and cuts the axes in A, B, C. Show that the locus of centre of the sphere OABC is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$ | 5 | CLO1 | U |

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International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)
Mid Term Examination, Spring-2026
Course Code: CSE-1223 Course Title: Discrete Mathematics
Total marks: 30 Time: 1 hours 30 minutes
[Figures in the right margin indicate full marks]

	Questions	CLO	Marks
1.a	Write the difference between Roster Method and Set Builder Notation with example.	1	2
1.b	Define Set Equality with some examples. Let $A = \{1, 2, a\}$ and $S = \{x, y, z\}$. Find the Cartesian product $A \times S$. What is the cardinality of $P(A \times S)$.	2	3
1.c	In a Cyber security workshop of 80 professionals, the following data was collected: 40 know Network Security, 35 know Cloud Security, 30 know Cryptography, 15 know both Network and Cloud Security, 12 know both Cloud Security and Cryptography, 10 know both Network Security and Cryptography, 5 know all three fields. Find the number of professionals who know: i) Only Cryptography. ii) Network and Cloud Security but not Cryptography. iii) Exactly two of the three fields. iv) None of the three fields. v) At least one of the three fields.	2	5
2.a	Write down the importance of quantification with example.	2	2
2.b	Define converse, inverse, and contrapositive of $(p \rightarrow q)$. Find the converse, inverse, and contrapositive of "It raining is a sufficient condition for my not going to town."	2	3
2.c	Let the domain be the set of all robots in BMTF factory. Also, $R(x)$: "x is a Repair robot.", $F(x)$: "x is Functional." and $W(x,y)$: "x Works with y." Now, Translate the following statements into logical expressions using the provided predicates: i) "All repair robots are functional." ii) "There is a robot that is functional and works with every repair robot." iii) "No repair robot is functional if it does not work with at least one other robot." iv) "If a robot works with all functional robots, then it is a repair robot." Or Consider the following predicates: $R(x)$: "x is a Repair robot.", $F(x)$: "x is Functional." and $W(x,y)$: "x Works with y.". Given domain be the set of all robots in BMTF factory, translate the following quantifier expressions into English statements based on the given predicates: i) $\exists x(R(x) \wedge \neg F(x))$ ii) $\forall x(F(x) \rightarrow \exists y(R(y) \wedge W(x,y)))$ iii) $\neg \forall x(R(x) \rightarrow F(x))$ iv) $\forall x \forall y(W(x,y) \rightarrow W(y,x))$	2	5
3.a	Differentiate between one-to-one and onto function with example. Is it possible for a function be both a time? Justify. Or Let f be the function from $\{a,b,c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible and if so what is its inverse?	2	2
3.b	Let h be a function from the set $S = \{a, b, c\}$ to itself such that: $h(a) = c$, $h(b) = a$, $h(c) = b$ also k be another function from the set S to the set $T = \{100, 200, 300\}$ such that: $k(a) = 300$, $k(b) = 200$, $k(c) = 100$. i) Find the composition $k \circ h$ by specifying the image for each element in the domain. ii) Determine if the composition $h \circ k$ is defined. Provide a valid argument to justify your answer based on the domains and codomains of the functions. Or Let f and g be functions from the set of integers to the set of integers defined by $f(x) = x + 2$ and $g(x) = 3x + 1$. What is the composition of f and g , and also the composition of g and f ? Draw $g \circ f$ and $f \circ g$ for $x = \{1, 2, 3\}$.	2	3
3.c	For each of relations (R_1, R_2, R_3, R_4) on the set $\{1, 2, 3, 4\}$ and R_5 on integers, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive. i) $R_1 = \{(2,4), (3,2), (3,3), (3,4), (2,2), (2,3)\}$ ii) $R_2 = \{(2,2), (3,3), (4,4), (1,1), (1,4), (4,1)\}$ iii) $R_3 = \{(2,4), (4,2), (2,1), (4,1)\}$ iv) $R_4 = \{(1,1), (2,2), (3,3), (4,4)\}$ v) $R_5 = \{(a,b) \mid a \leq b\}$	2	5

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 International Islamic University Chittagong
 Department of Computer Science & Engineering
 B. Sc. in CSE Mid Term Examination, Spring-2026
 Course Code: PHY-1201 Course Title: Physics-II
 Total marks: 30 Time: 1 hours 30 minutes

[Answer all the questions. Figures in the right hand margin indicate full marks.]

1.
 - a) Define electric field strength. CLO1 R 1
 - b) For a circular turn of wire, if "a" is the radius of the wire and "b" is the distance between the center and the point of interest, then obtain an expression for the electric field at a point on the axis of a charged circular turn of wire for $b \gg a$ in detail with proper figure. CLO2 U 6
 - c) A positive charge of $q_1 = 2.5 \times 10^{-7} \text{ C}$ is placed at a distance of 0.15m from another positive charge of $q_2 = 8.5 \times 10^{-7} \text{ C}$. (i) Find the electric fields of q_1 and q_2 . (ii) Find the point on the line joining them where the electric field becomes zero. CLO3 A 3

 2.
 - a) State Biot-Savart law. CLO1 R 1
 - b) Using Biot-Savart law, derive an expression for the magnetic induction at a point due to a straight conductor carrying current in detail with proper figure. CLO2 U 6

OR

Define the coefficient of self-induction of a coil. Deduce a mathematical expression for the self-inductance of a solenoid having N turns in detail with proper figure.

 - c) A solenoid having a core with 2000 turns of cross-section 6 cm^2 , and is 30 cm long. Calculate (i) the coefficient of self-inductance, (ii) if the coil is made of half air and half iron (relative permeability 600), find the value of L. CLO3 A 3
-
- $$L = \frac{\mu_r \mu_0 n^2 A l}{1}$$
3.
 - a) Define RC Circuit. CLO1 R 1
 - b) Obtain an expression for the charging of a capacitor connected to a D.C. battery source with a series resistor, with a detailed figure and charge growth curve. CLO2 U 5
 - c) A $70 \mu\text{F}$ capacitor initially uncharged is connected through a 300-ohm resistor to a 12 V battery. Calculate: CLO3 A 4
 - (a) The magnitude of the final charge q_0 on the capacitor.
 - (b) Time constant of the circuit.
 - (c) The time after which the capacitor will be charged to $\frac{1}{2} q_0$, when it is connected to the battery.
 - (d) The time after which the capacitor will be charged to $0.80 q_0$, when it is connected to the battery.

OR

A square coil of side "d" carries a current "i". Calculate the magnetic induction at the centre of the coil.

International Islamic University Chittagong
Department of Computer Science and Engineering

Mid Term Examination, Spring-2026

Program: B.Sc. Engineering in CSE

Course Code: EEE-1221

Course Title: Electronics

Time: 1 hour 30 minutes

Full Marks: 30

[Answer all the questions from the followings. Figures in the right margin indicate full marks]

Course Outcomes (COs) and Blooms Levels are mentioned in additional columns.

Marks

- 1) a) Suppose an electronic device requires the same output polarity for any polarity of the ac input. Design a rectifier using four diodes to get the required output and derive the efficiency of a full-wave rectifier CO1 Ap 5
- 1) b) Fig.1a and Fig.1b show the center-tap and bridge type circuits having the same load resistance and transformer turn ratio. The primary of each is connected to 230V, 50Hz supply. CO2 Ap 6
- i) Find the d.c voltage in each case ii) PIV for each case for the same d.c output.

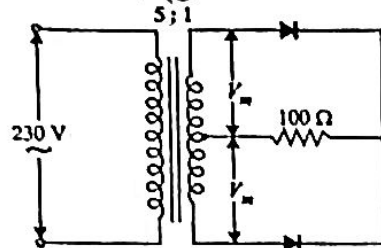


Fig.1a

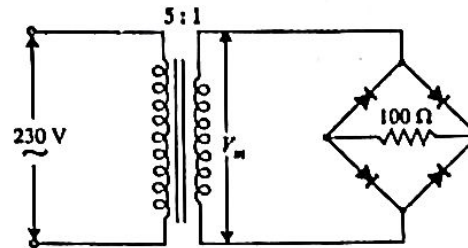


Fig.1b

OR

- 1) b) A crystal diode having internal resistance $r = 13 \Omega$ is used for half-wave rectification. If the applied voltage is $v = 57 \sin(\omega t)$ and the load resistance is 999Ω , find: CO2 Ap 6
- (i) a.c. power input and d.c. power output
- (ii) efficiency of rectification
- 2) a) Draw and explain the input and output Characteristics of Common-Base Configuration and find the relationship between α and γ . CO1 U 4
- 2) b) Determine whether or not the transistor in Fig. 2b is in saturation. Assume $V_{knee} = .2V$. CO2 An 6

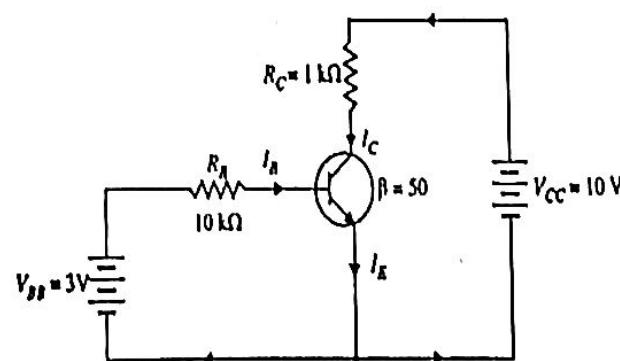


Fig.2b

- 3) a) What is biased diode clipping circuit? Design and explain the output waveform for a positive biased diode clipping. CO1 Ap 5

- 3) b) Define Clamper Circuit. Sketch the output waveform for the circuit shown in Fig. 3a. It is given that discharging time constant (CRL) is much greater than the time period of input wave: Fig 3a

CO2 E 5

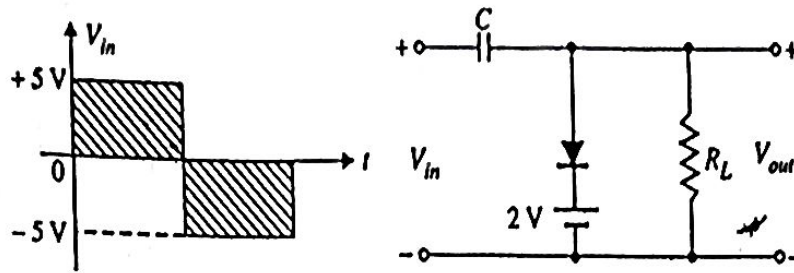


Fig 3a

- 3) b) Find the output voltage and draw the output waveform for the following circuit (Fig. 3b) using given input. Consider the diodes are ideal.

e CO2 E 5

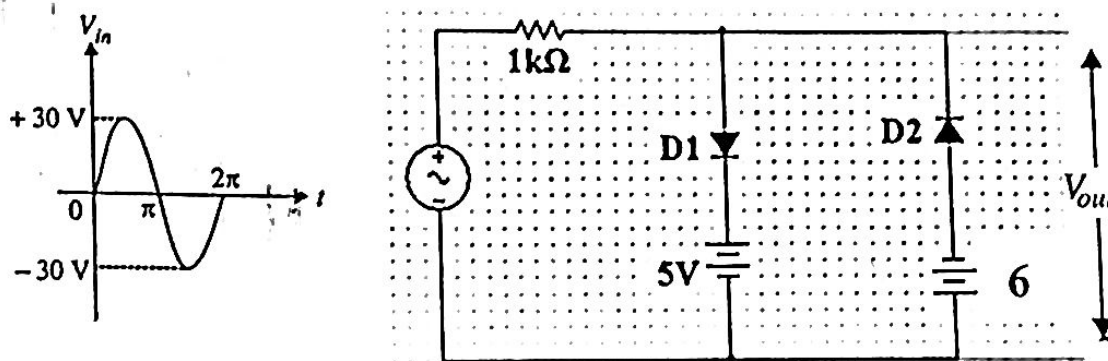


Fig.3b